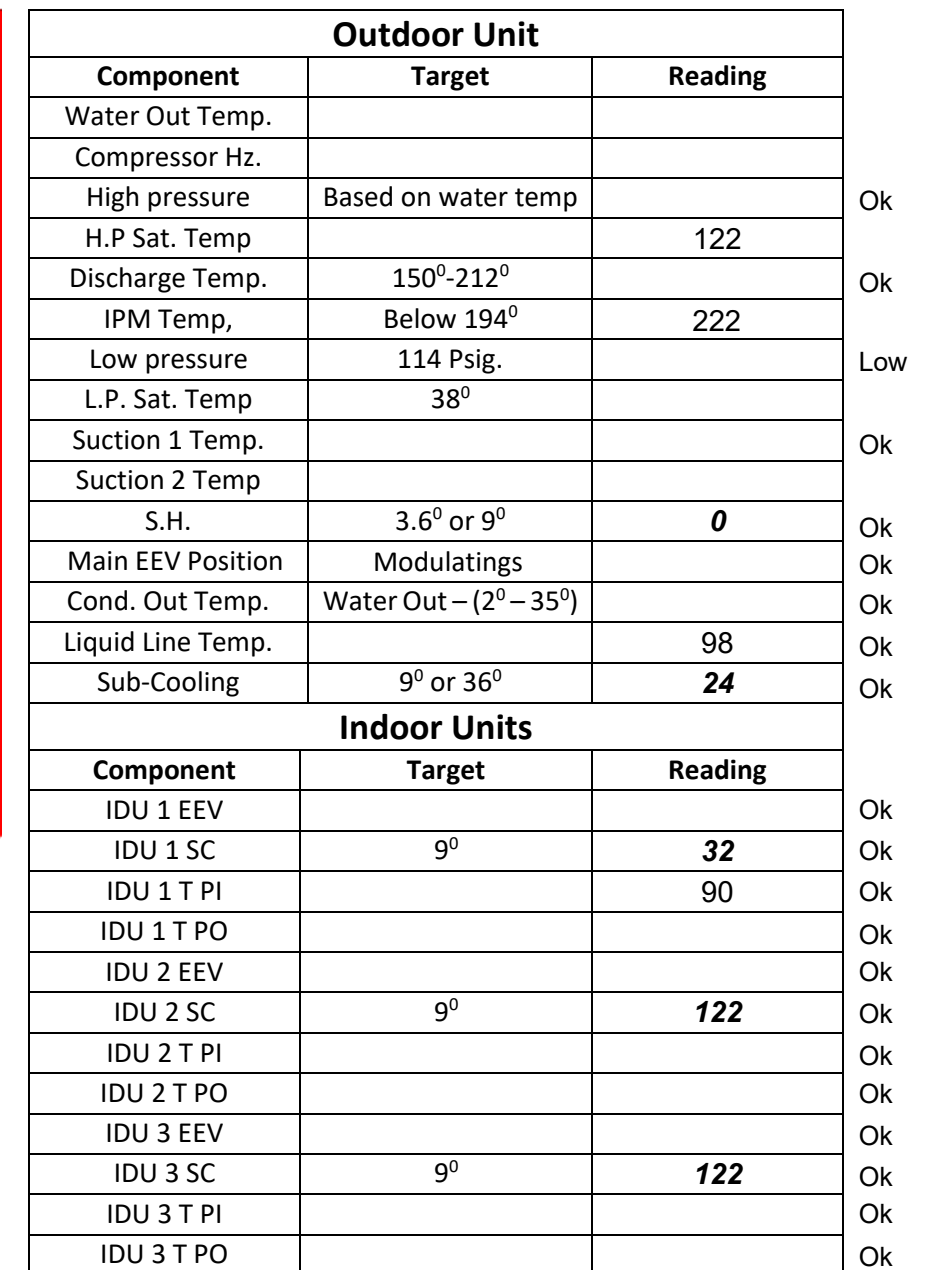




Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.



DVM S 3 Phase Water Heat Pump
Heating Target Operation

Water Source Outdoor Unit				
Component	Abbreviation		Heating Mode	Comments
Compressor	Comp	Operation	Target High pressure control (default: 406 psig)	Primary function is to maintain head pressure (range 356-455psig)
		Range	20~160Hz	
Main EEV	E_M	Operation	HR - Target SH = Water temperature - 5.4°F	Valve modulates to maintain Evaporator Superheat
		Range	125 - 2000 Steps	Suction Superheat 1.8°F - 12.6°F
EVI EEV	E_EVI	Operation	On (Open)	By-Pass Mode 18°F Superheat Between pipe in and pipe out
		Range	0-480step	EVI Vapor Injection Mode 3.6°F Superheat Between pipe in and pipe out
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 50.4 or 50Hz↓	By-pass mode, Discharge Temperature control
			Off (Close): T_W < 45°F ↓ and 65Hz↑	Low Ambient heating Operation Vapor Injection circuit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open)	Off during start up if DSH is < 18°F. Off for 2 min. if DSH is < 18°F On when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	
Liquid Line Solenoid Valve	V_L	Operation	On (Open)	
Main Cooling Solenoid	V_MC	Operation	Off (Closed)	
Hot Gas Charging Solenoid Valve	V_HC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the water out temperature	Low Side saturation temperature is <i>"typically"</i> 2-10 degrees lower than water temperature
Indoor Unit IDU				
Component	Abbreviation		Heating Mode	Comments
Room Temperature	Room Temp.	Operation	61-86 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub_cooling Control	Used to calculate Evaporator Sub-Cooling (91°F - 122°F)
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil (113°F - 194°F)
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
			Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	